

Intern Feedback Sentiment Analysis Using Machine Learning and Transformers

A Report on Sentiment Classification of Intern Feedback

Submitted by:

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This report presents the methodology, results, and conclusions of a sentiment analysis task performed on intern feedback data as part of an internship project.

1. Title

Intern Feedback Sentiment Analysis Using Machine Learning and Transformers

2. Objective

Analyze intern feedback to classify sentiments as **Positive, Negative, or Neutral** and identify areas where intern satisfaction can be improved.

3. Dataset

- Total records: **1,500**
- Features/columns: **8**
- Text column: **Feedback_Text**
- Target column: **Sentiment**
- Missing values: **None**
- Positive: **520 (34.67%)**
- Negative: **500 (33.33%)**
- Neutral: **480 (32.00%)**

4. Data Preprocessing

The following preprocessing steps were carried out before model training:

- Loaded the dataset into the working environment.
- Checked the dataset for missing values.
- Selected the feedback text and sentiment columns for analysis.
- Split the data into **80% training** and **20% testing** sets.
- Used stratified splitting to maintain the original class distribution across both sets.

5. TF-IDF Feature Extraction

Term Frequency-Inverse Document Frequency (TF-IDF) was used to convert the raw feedback text into numerical features that machine learning models can process. Each feedback entry was transformed into a vector of weighted word importance scores.

Dataset	Shape (Samples, Features)
Training	(1200, 633)
Testing	(300, 633)

6. Model 1 — Logistic Regression

A Logistic Regression classifier was trained using the TF-IDF features extracted from the feedback text, with the parameter **max_iter=1000** to ensure convergence.

Results:

- Accuracy: **99%**
- Macro F1-score: **0.99**

Confusion Matrix:

	Pred: Negative	Pred: Neutral	Pred: Positive
Actual: Negative	99	1	0
Actual: Neutral	0	96	0
Actual: Positive	0	2	102

7. Model 2 — Pretrained Transformer

Model used: RoBERTa — **cardiffnlp/twitter-roberta-base-sentiment-latest**

This model was **pretrained** and used directly for sentiment prediction without any additional fine-tuning on the intern feedback dataset.

Results:

- Accuracy: **77%**
- Macro F1-score: **0.76**

Confusion Matrix:

	Pred: Negative	Pred: Neutral	Pred: Positive
Actual: Negative	91	9	0
Actual: Neutral	3	51	42
Actual: Positive	0	15	89

8. Model Comparison

Model	Accuracy	Macro F1
Logistic Regression	0.99	0.99
Pretrained Transformer	0.77	0.76

9. Satisfaction Analysis

The main areas identified from negative and neutral feedback are as follows:

- Mentor communication and guidance
- Meaningful and less repetitive tasks
- Learning and skill-development opportunities
- Career development
- Training quality and challenge level
- Workplace pressure

10. Transformer Limitation

The Transformer was not trained from scratch because the dataset contained only 1,500 records and the available laptop environment was CPU-based. Training a Transformer from scratch would require substantially more training data and computational resources. Therefore, a pretrained Transformer was used as a practical baseline.

It is also important to note that **not fine-tuning the pretrained model is a significant limitation** of this approach, and may explain its comparatively lower performance.

11. Final Conclusion

In this task, intern feedback was analyzed using sentiment classification to understand overall intern satisfaction and identify areas for improvement. The dataset contained 1,500 feedback records, consisting of 520 positive, 500 negative, and 480 neutral reviews.

Two approaches were evaluated. Logistic Regression with TF-IDF achieved an accuracy of 99% and a macro F1-score of 0.99. A pretrained RoBERTa Transformer, used without fine-tuning, achieved an accuracy of 77% and a macro F1-score of 0.76. The Transformer showed more difficulty distinguishing neutral feedback, while Logistic Regression performed better on this particular dataset.

The Transformer was not trained from scratch because the dataset was relatively small and the available laptop environment was CPU-based. Training a Transformer from scratch would require substantially more data and computational resources. Therefore, a pretrained Transformer was used as a practical baseline.

Analysis of negative and neutral feedback highlighted several areas that could improve intern satisfaction. These included better mentor communication and guidance, more meaningful and less repetitive tasks, greater opportunities for learning new tools and skills, career-development guidance, more engaging training, and better management of workplace pressure.

Overall, the sentiment analysis provided useful information about intern experiences and highlighted specific areas that organizations can consider when improving their internship programs.